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The Exploration of a Low-Intensity Paraprofessional-Delivered Peer Training in Inclusive High School Classrooms

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Abstract

Social competence is an essential skill that can be promoted in inclusive high school classrooms. For high school students with developmental disabilities, peer-mediated interventions have shown promise for increasing social communication with feasible implementation. This study focused on increasing the appropriate peer interactions of three high school students with autism or intellectual disabilities in inclusive general education classrooms. We used a single-case multiple probe design to evaluate the effectiveness of a low-intensity peer-mediated intervention delivered by a paraprofessional to six peer partners. Our visual analysis of the data resulted in determining a functional relation between the intervention and students' appropriate peer interactions. The peer partners and educators found the goals, procedures, and outcomes to be beneficial, important, and practical. We also discuss the implications, limitations, and ideas for future research.

Keywords

paraprofessional, peer training, social communication, inclusion, high school

For adolescents and young adults, developing social communication competence is among the most crucial educational outcomes before their high school graduation (Rubin et al., 2011). Social communication skills, ranging from greetings, information sharing, advocating, to connecting with others, are shown to be associated with improved academic learning, behavioral outcomes, employment opportunities, and quality of life. For many high school students with developmental disabilities (DD; e.g., autism or intellectual disability [ID]), learning to master these critical skills is often an educational priority and part of their Individualized Education Programs (IEPs; Wagner et al., 2005). Parents reported their adolescents with DD often struggled to make friends and engage in peer interactions (Wagner et al., 2005). Given that placement in a general education classroom alone does not result in naturally occurring peer interactions for high school students with DD (Chung et al., 2019; Herbert et al., 2020), educational teams are facing the tasks of selecting and implementing effective and feasible strategies to support social communication of students with DD across inclusive environments.

Among the strategies and practices for promoting social communication, peer-mediated interventions have shown promise for the substantiality and feasibility of implementation and for benefiting high school students with and without DD (Kamps et al., 2017; Schaefer et al., 2016). While researchers have shown to be effective intervention agents for peer-mediated interventions for high school students (e.g., Herbert et al., 2020; Hughes et al., 2013), another line of peer-mediated interventions has focused on equipping practitioners—specifically paraprofessionals—to deliver peer training in high school general education classrooms (Carter et al., 2016; Huber et al., 2018). Carter and colleagues (2016) examined the impacts of peer support on 51 high school students with DD. Following an average of 2.5-hr initial training, paraprofessionals and special educators attended another 45-min meeting to develop specific plans. The practitioners then delivered a 30-min training to peers and started providing support in classes. As a result, the students increased their peer interactions and expanded their social networks. The practitioners reported the time required was feasible. Using a less time-intensive intervention package, Huber et al. (2018) explored factors that impacted the effect of peer support for three high school students with DD. The researcher provided an initial 1.5- to 2-hr paraprofessional training. Each paraprofessional then recruited peers and delivered a 15- to 25-min peer training session. Upon establishing the effectiveness of initial peer supports, researchers manipulated specific environmental variable(s)

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Table 1. Information for Student Participants in the Peer Training Study

Characteristic	History	Food Science A	Food Science B
Focus student	Jacob, 21 years old, 12th grade	Max, 17 years old, 10th grade	Barry, 19 years old, 12th grade
Diagnosis[a]	Visual impairment and ID	Autism	ID
% of Inclusion	< 40% of the school day	40%–79% of the school day	40%–79% of the school day
VABS-II[b]	55/64/57 (low/low/low)	60/59/57 (low/low/low)	55/64/56 (low/low/low)
IEP goals[c]	Asking for clarifications as needed	Initiating a greeting, answering questions, and making requests	Taking turns, initiating, and using appropriate personal space
Peer partners	Bryan, 17 years old, 11th grade; A+ grade	Morgan, 18 years old, 12th grade; B- grade	Austin, 18 years old, 12th grade; D grade
	Molly, 17 years old, 11th grade; A+ grade	Maya, 16 years old, 11th grade; A+ grade	Rae, 17 years old, 12th grade; No grade[d]

Note. ID = intellectual disability; VABS-II = Vineland Adaptive Behavior Scales: Second Edition; IEP = Individualized Education Program.
[a]Educational diagnosis. [b]Social, communication, and daily living skills domain scores and adaptive levels. [c]Relevant social communication goals. [d]This peer attended the class for no credits.

and delivered one brief training to peers and paraprofessionals to implement adapted peer supports. As a result, one of the three students increased his responses with peers. When asked to rate whether the time required for the intervention was reasonable, paraprofessionals' responses ranged from neutral, agree, to strongly agree.

With various intensity and intervention agents of peer-mediated interventions that have been explored for high school students with DD, we are interested in continuing this line of work with two foci: (a) equipping paraprofessionals for peer-mediated interventions in high school classrooms and (b) experimenting with low-intensity training (45–60 min for paraprofessionals and 3–5 min for peers with simple directions requiring a minimal additional effort). We chose these aims for two main reasons. First, there remains to be a gap among paraprofessional training and peer-mediated interventions in high school classrooms (Brock & Anderson, 2021; Watkins et al., 2015). While paraprofessionals are among the key personnel supporting students with DD across grades and classrooms (Giangreco, 2021), more intervention studies have involved elementary school students or were conducted in segregated and/or non-classroom settings. Watkins et al. (2015) examined peer-mediated intervention studies published from 2008 to 2014 that targeted peer interactions of students with autism in general education settings. Among the 14 studies identified, only three were conducted in high schools. Brock and Anderson (2021) identified 36 studies published from 2012 to 2019 that involved training paraprofessionals of students with DD. In five studies, paraprofessionals delivered strategies in high school general education classrooms, with only two peer-mediated interventions conducted by one research team across two different geographical locations (i.e., Carter et al., 2016; Huber et al., 2018).

Second, as teacher educators, we are especially interested in exploring effective paraprofessional-delivered peer-mediated intervention that requires less training time and effort. Educators continue to face various challenges when selecting and implementing effective and feasible interventions in their classrooms (Long et al., 2016; Neal et al., 2018). Long et al. (2016) distributed a statewide survey to examine a total of 12,19 teachers' experiences and perceptions on intervention implementation. Time and duration were among the most common barriers reported and identified as the most challenging to address by teachers across grade levels. While the effectiveness of peer-mediated interventions has been demonstrated with intensity varied across planning, duration, and activities involved, it is important that researchers continue to explore and experiment with the intensity of interventions, including training time and efforts required for educators. This way, educational teams have options of effective strategies to select from based on the intended student outcomes and available time and resources. Thus, we developed our study to answer the following research questions: RQ1. Is there a functional relation between a low-intensity paraprofessional-delivered peer training and appropriate peer interaction of high school students with DD in inclusive general education classrooms? RQ2. How will students, peers, and educators perceive the social validity of the intervention?

Method

Participants

Students and Peers. Three focus students and six peer partners participated in this study (see Table 1). All were European Americans, except Max, who is a Chinese American. A focus student was included if they (a) had a diagnosis of autism or ID, (b) attended at least one inclusive general education class at a high school, and (c) had communication needs as reported by their special educators and in their IEPs. After consulting with the classroom teachers, the paraprofessional selected two peer partners for each focus

student based on existing interaction, prior relationships, and/or personal characteristics (e.g., outgoing, caring, kind).

Jacob was a hard worker who enjoyed country music. He wore glasses and had visual acuity of light perception in one eye. Jacob used a cane for mobility and navigated independently. He enjoyed initiating greetings and carrying on conversations with peers and adults, using complete sentences. Bryan, a peer partner, already sat in proximity to Jacob and regularly interacted with Jacob and another student with DD in history class prior to the study. The history teacher recommended Molly, a sociable classmate, as another peer partner.

Max was a talented young man who played violin. He memorized his schedule and independently went to his classes. Max primarily used English with some Chinese words to communicate and had delayed and immediate echolalia in both languages. Max responded to greetings and some basic personal questions with phrases but rarely initiated. Max often required prompts to answer social questions (e.g., what did you do last night?). At the time of the study, Max carried a tablet with a communication application but used speech as his primary communication mode. Prior to the study, Morgan was in Max's assigned group in their food science class and appeared to be friendly, so he became a peer partner. Maya, another peer partner, had experience interacting with another student with a disability before the training.

Barry was very social and liked to sing. He enjoyed being around his peers and greeting his friends in the hallways, as he navigated the school environment independently. He communicated well during routine interactions using short phrases and sentences, with scripted responses and echolalia. Sometimes he would talk in an inappropriately loud voice. Austin, a peer partner with a learning disability, was familiar with Barry and interacted with him in another class. Prior to the study, Rae was part of the school's existing peer partner program and attended the class as Barry's peer partner for no school credit.

Paraprofessional and Teachers. After sharing the purpose of the study, we asked the special educators to identify students who met our inclusion criteria. We then invited the paraprofessional, Angela, an European American who supported the three focus students in their general education classrooms to serve as the interventionist. She has an associate degree and held a paraprofessional license that required at least 60 hr of coursework. With 19 years of experience at the school, Angela was passionate about supporting students in inclusive classes and would seek opportunities to advance her knowledge and skills. She reported attending two workshops on promoting peer interactions prior to the study. Two special educators and three general education teachers, all European Americans with a range of 2 to 27 years of experience, also participated in the study by completing a post-intervention survey.

Settings

We conducted the study in a rural Midwestern high school with an estimated 1,000 students and a racial/ethnic distribution of 92% European American, 5% two or more races/ethnics, 3% Latinx, 1% African American, and 1% Asian. About 15% of the students received free or reduced-price meals and 10% of the students had IEPs. Across observations, Jacob's history, Max's food science, and Barry's food science classes had an average of 17, 22, and 17 students, respectively. Jacob's history class had the traditional seating arrangement of individual desks in rows. Max's and Barry's food science classes were in the same room at different times. There were individual desks in the center, surrounded by a few large tables and several kitchen stations—each equipped with a sink, an oven, and a stove. The average observation duration in each class was 45, 41, and 44 min in history, food science A (Max), and food science B (Barry), respectively. Each focus student completed work alongside their peers after receiving instructions from the general education teacher. The paraprofessional assisted the focus students when necessary.

Data Collection

We used direct observations with data sheets to measure student interaction and classroom variables. Specifically, we used 1-min partial interval recording for interaction variables and momentary time sampling for classroom variables. Each observer wore an *Invisible Clock* that vibrated at the beginning of each minute. We also used event recording to record the number of prompts provided by peer partners and the number of different peers interacting with the focus student during each observation. The researchers, both prior special education teachers, served as observers. The first author, the primary observer, met with the secondary observer (the second author) to discuss coding and protocols. They practiced coding multiple videos showing interactions until reaching a 90% agreement. Below are definitions of our measures.

Student Interaction. We selected *appropriate peer interaction* as our primary dependent variable. Appropriate peer interactions included age-appropriate and socially acceptable initiations and responses with peer partners and other peers. We defined *initiations* as verbal and/or nonverbal behaviors emitted by the focus student with clear communication intent to evoke a response from a peer partner, another peer without disabilities, or an adult. Examples of initiations may include the student saying hello to a peer or touching a peer's shoulder for attention. Non-examples of initiations

may include the student passing a worksheet to a peer or producing self-stimulated vocalizations. We coded a new initiation when 5 s elapsed between two interactive behaviors or when a new topic or a new partner was introduced. We defined *responses* as verbal/nonverbal behaviors demonstrated by the student that overtly or passively acknowledged an initiation or response from a peer partner, another peer, or an adult (e.g., answering a question verbally or with a head nod). Nonexamples of responses may include receiving a worksheet handed by a peer or following directions given by a teacher. A response was produced within 5 s of an initiation or a previous response by a peer or adult. We defined inappropriate interactions as any initiation or response toward a peer partner, another peer, or an adult that was not typical for high school students in similar situations (e.g., inappropriate volume or sounds, prolonged hugs).

To measure appropriate peer interaction, we totaled the number of intervals where the focus student initiated or responded toward a peer (i.e., a peer partner or another peer without disabilities) and minus the intervals where the focus student had an inappropriate initiation or response. For each initiation and response, we recorded whether it was independent (i.e., the student produced the behavior spontaneously) or prompted (i.e., the student produced the behavior following a verbal or nonverbal prompt from a peer or an adult). To capture sustained interaction exchanges, for each interval, we recorded if the focus student took three or more communicative turns (i.e., one initiation and two responses, or three consecutive responses) with a peer partner, another peer, or an adult. We also counted the number of prompts provided by a peer partner with an intention for the focus student to initiate or respond to another peer or adult as well as the number of peers who received initiations or responses from the focus student.

Classroom Variables. At the beginning of each interval, we recorded two classroom variables: the instructional format and proximity. The instructional format received by the focus student was coded as (a) large group (i.e., with seven or more classmates), (b) small group (i.e., with one to six classmates), (c) independent work, (d) no instruction (e.g., free time), or (e) gone (i.e., the student was not in the room). We also coded if the focus student was in proximity (about three feet) to their peer partners, other peers, or the paraprofessional.

Research Design

We used a single-case multiple probe design across students and classrooms to evaluate the impacts of the paraprofessional-delivered peer training on students' appropriate peer interactions (Ledford & Gast, 2018). We controlled threats to internal validity and established experimental control through concurrent baselines, repeated observations, and staggered interventions. We used a multiple probe design with intermittent baseline data collection to decrease observer reactivity and for practical time management. During baseline, we collected an initial three to four probes continuously. For the second and third tiers, we collected additional probes right before the introduction of the intervention (a lapse of 2–3 weeks). During the intervention phase, we collected data continuously for the remainder of the school year, except for the final intervention data point for the first tier due to student absence (a lapse of 10 days). We selected appropriate peer interactions (prompted and independent initiations and responses) as the primary dependent variable to account for variations in the focus students' communication skills and support needs. Following a baseline phase with at least four sessions of contratherapeutic data, we trained the paraprofessional who then trained the peer partners of the focus student in the first tier. After the first focus student demonstrated increased appropriate peer interactions with a change in level or trend over at least three consecutive data sessions, as compared with the baseline data (based on visual analysis, not percentage), we introduced the training and intervention data collection to the second tier, which had established a stable baseline data trend. We used the same process for the third tier. To determine if there was a functional relation between the paraprofessional-delivered peer training and the focus students' appropriate peer interactions, we used visual analysis to evaluate changes in levels, trends, and/or variability between baseline and intervention data (Ledford & Gast, 2018). We also calculated Tau-U to estimate the intervention effects using a nonparametric measure of nonoverlapping data points and trends between baseline and intervention (Vannest & Ninci, 2015). We discussed Tau-U in terms of having a small effect ($<.20$), moderate effect (.20–.60), large effect (.60–.80), or very large effect ($>.80$; Vannest & Ninci, 2015).

Procedures

Baseline. Upon receiving permissions, we began observations and asked the classroom teachers and the paraprofessional to continue their typical instruction and activities. We observed each focus student and collected data on all interaction and classroom variables. We started each observation when the bell rang at the start of class or when the focus student entered the room (whichever came first), and ended each observation when the bell rang to change classes. During data collection, the observer stood by the side of the classroom and kept an appropriate distance optimal for observation of the focus student (e.g., a distance that allowed for reliable coding of the student's interactions with visual and audio inputs). We collected and monitored baseline data for a stable or contratherapeutic trend. We began the intervention with Jacob first because we

hypothesized that Jacob and his peer partners would respond well to the intervention and his baseline data showed a contratherapeutic trend. Due to the school semester time restraints, we were only able to collect four baseline data points for the first tier (Jacob's history class).

Paraprofessional Training. After baseline data collection for each focus student, the second author (a university-based researcher) met with the paraprofessional in an empty classroom or the teacher's lounge before or after school. Three individual meetings—each lasted 60, 45, and 45 min for Jacob's, Max's, and Barry's classes—occurred in a didactic format. During each meeting, the researcher provided the paraprofessional with a peer training handout (available upon request) and discussed (a) the purpose of the project, (b) the focus student's interests and hobbies (as shared by the paraprofessional), (c) the course content and topics, (d) logistics of peer training, and (e) strategies and information to share with peer partners to increase appropriate interaction of the focus student.

We adapted these strategies based on prior peer support research (e.g., Carter et al., 2011, 2016), including asking the focus student questions at three different times and prompting the focus student to ask other classmates questions at three different times throughout the class period, modeling how to initiate and respond to other students when appropriate, introducing the focus student to other students when possible, and inviting the focus student to join them when moving around. As part of the peer training handout, the researcher included a list of class-related and social conversational topics and phrases that peers and the student could say throughout the class (e.g., "How is your project going?" "How was the game last night?"). At the end of the meeting, the researcher reminded the paraprofessional to deliver training to peer partners during the next class session, monitor student interactions following the training, reduce her proximity to the focus student, and prompt peer partners if needed. We asked her to continue her typical support in the classes that had not yet received the intervention.

Peer Training. Following paraprofessional training, the paraprofessional delivered one 3- to 5-min session of peer training by talking to both peer partners at the side of the classroom prior to or at the beginning of class. Using the peer training handout as a guideline, the paraprofessional discussed the project purpose and the same strategies (as described in Paraprofessional Training section above). The paraprofessional also shared additional information and examples based on each focus student's interests, support needs, and class activities. For example, the paraprofessional shared Jacob's recent rock-climbing accomplishment. For Barry, the paraprofessional discussed how to redirect his hugs with more appropriate behaviors (e.g., fist bumps or handshakes). At the end, she emphasized two reminders (asking the focus student questions at least three times and prompting him to talk to other classmates at least three times throughout the class at appropriate times) and the list of class-related and social conversation ideas and phrases.

Intervention. Following peer training, we collected intervention data on all variables in the same manner as baseline observations. The paraprofessional monitored the focus student's interactions daily and made sure at least one peer partner was in proximity to the focus student during the class (e.g., inviting a peer partner to work with and talk to the focus student). She provided intermittent feedback to the peer partners during and/or at the end of the class. The intervention continued for each class until the end of the school year. For the last tier, we were only able to collect four intervention data points with Barry.

Reliability and Fidelity

For each student and class, the secondary observer joined the primary observer and independently collected data on all variables during at least 20% of baseline and 25% of intervention observations. We calculated interobserver agreement (IOA) by using point-by-point agreement (i.e., student interaction and classroom variables) and total percentage of agreement (i.e., peer partner prompts and different peers interacted). If IOA was below 80% for a variable, both observers discussed the discrepancies, reviewed the coding definitions, and reached a consensus. We also evaluated the procedural fidelity for paraprofessional and peer training using a checklist to record items discussed. If procedural fidelity was below 90%, additional feedback would be provided to the paraprofessional or peer partners during a follow-up session.

Social Validity

At the completion of the study, we interviewed the focus students and peer partners individually and asked the paraprofessional and teachers to complete a written questionnaire to share their perceptions on the goals, procedures, and outcome of the intervention (Snodgrass et al., 2018). Specifically, we asked the focus students four questions about their experience (e.g., "Do you like to talk to 'the name of peer partner'? Why? What do you talk about?"). We asked peer partners nine open-ended questions (e.g., "Was the information [training] helpful? Why or why not? What did you learn from this project?"). The social validity questionnaires for the paraprofessional and the teachers included 12 open-ended questions and 6 open-ended questions, respectively (e.g., "What [if any] changes have you noticed associated with the focus students, the peers, and the paraprofessionals as a result of this project?"). After

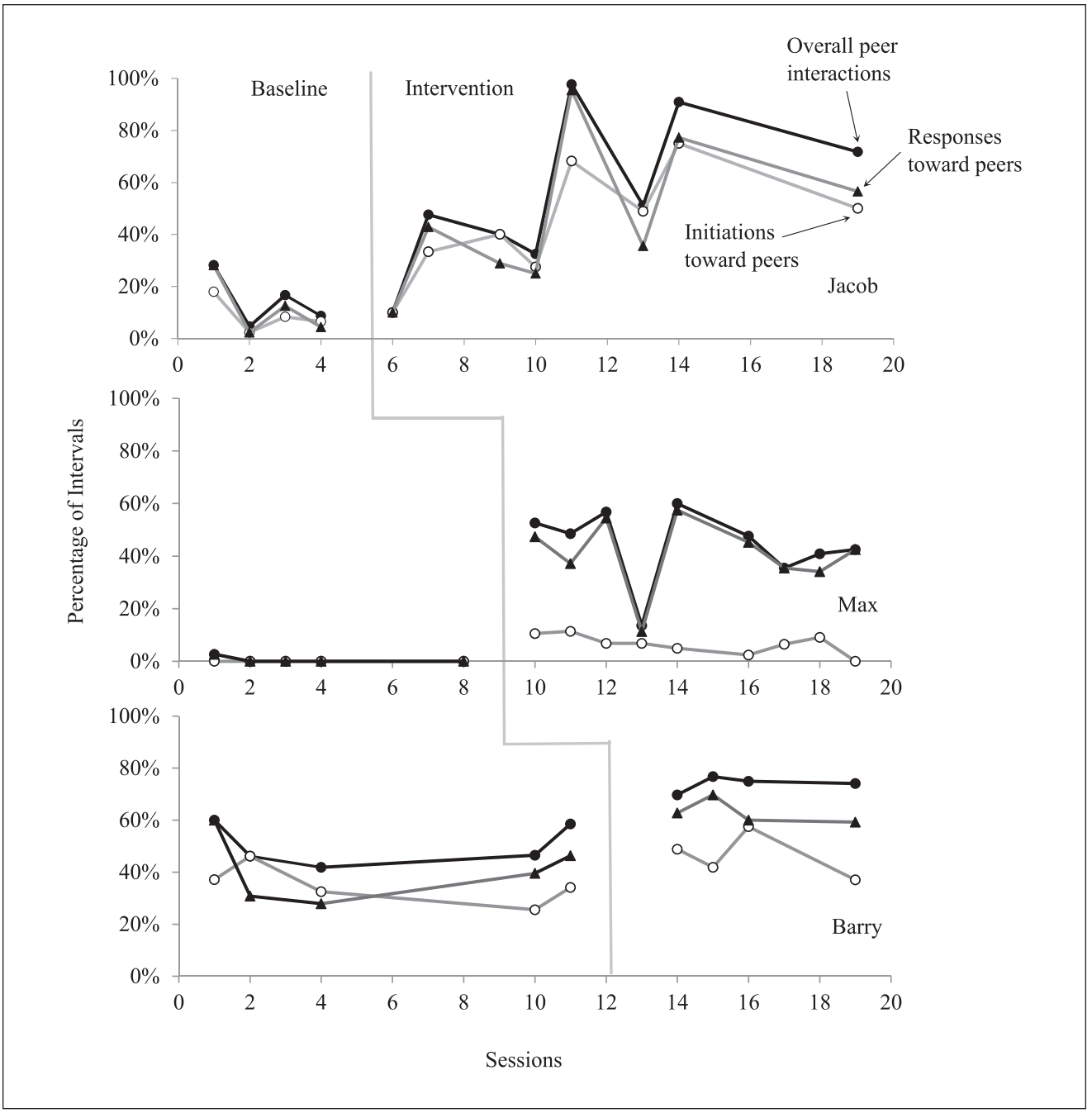


Figure 1. Appropriate peer interactions (independent and prompted initiations and responses).

transcribing interviews and questionnaires, we analyzed participant responses and identified themes related to the social importance and feasibility of the intervention.

Results

Student Interactions

Appropriate Peer Interaction. Figure 1 shows the percentage of intervals when the focus student had an appropriate interaction with a peer—either a peer partner or another peer in the general education classroom. In addition to the primary variable, we also graphed the student's appropriate initiations and responses toward a peer. During baseline, Jacob's overall peer interaction represented a variable descending trend at the medium-low to low level. Although his intervention data did not show an increase immediately after his peer partners received the brief training delivered by the paraprofessional, changes in level, trend, and variability were noted with a highly variable ascending trend

Table 2. Student Interactions and Proximity by Condition.

	History (Jacob)		Food Science A (Max)		Food Science B (Barry)	
Measures	Baseline	Intervention	Baseline	Intervention	Baseline	Intervention
With peer partners						
Independent initiations	8 (2–18)	42 (3–73)	0 (0–0)	1 (0–7)	35 (26–54)	33 (19–52)
Independent responses	11 (2–26)	45 (8–93)	0 (0–0)	14 (0–14)	39 (30–47)	55 (45–63)
Prompted initiations	1 (0–3)	2 (0–5)	0 (0–0)	5 (0–11)	0 (0–0)	3 (0–8)
Prompted responses	0 (0–0)	0 (0–2)	0 (0–0)	28 (0–50)	0 (0–0)	3 (0–11)
Appropriate interactions[a]	13 (4–26)	54 (8–98)	0 (0–0)	41 (14–60)	39 (21–60)	62 (48–85)
With other peers						
Independent initiations	1 (0–4)	2 (0–8)	0 (0–0)	0 (0–0)	18 (3–39)	24 (19–28)
Independent responses	2 (0–4)	2 (0–8)	0 (0–0)	1 (0–7)	19 (6–27)	22 (9–35)
Prompted initiations	0 (0–0)	0 (0–2)	0 (0–0)	1 (0–5)	0 (0–0)	3 (0–8)
Prompted responses	0 (0–0)	0 (0–0)	1 (0–3)	2 (0–7)	1 (0–3)	1 (0–2)
Appropriate interactions[a]	3 (0–9)	2 (0–8)	0 (0–0)	4 (0–11)	18 (9–29)	34 (19–49)

Note. Mean (range) percentage of intervals in which the focus student was present in the class.
[a]Calculated by total intervals with an initiation or a response minus the number of intervals with an inappropriate interaction.

toward the high level. His Tau-U effect size was 0.875 $CI_{90\%}$ [.27, 1.00] ($p = .017$), indicating a very large effect. Jacob's peer initiation and response data followed a similar data path as his overall peer interaction data in both baseline and intervention phases.

Max had a very low-level stable baseline data for overall peer interactions. After the intervention, he had a substantial immediate increase (from 0% to 53%) in his appropriate peer interactions. With some variabilities, most of his intervention data were maintained at the medium to medium-high level. Only one intervention session was at the medium-low level but was still higher than his baseline data. His Tau-U effect size was 1.0 $CI_{90\%}$ [.45, 1.00] ($p = .002$), also indicating a very large effect. His response data followed a similar data path as his overall interactions, but his initiations toward peers were only slightly higher than baseline levels in all sessions except for the last session.

For overall peer interactions, Barry had somewhat stable baseline data at the medium to medium-high level. Following the intervention, his data showed a slight immediate increase (from 59% to 70%), followed by a stable trend at a high level. It should be noted that Barry was the only student who demonstrated inappropriate interactions across phases, and his inappropriate interactions toward his peer partners or other peers decreased from baseline to intervention, with a mean percentage of intervals of 18% (range: 14%–23%) to 7% (range: 2%–10%), respectively. Barry had a Tau-U effect size of 1.0 $CI_{90\%}$ [.33, 1.00] ($p = .014$), indicating a very large effect, although his Tau-U result should be interpreted cautiously due to the higher baseline level with an ascending trend. His response data illustrated an increase in level with more stable data in comparison to the baseline. For initiations, intervention data were variable at a slightly higher level than his baseline data. In summary, the results of the visual analysis for the three focus students indicated changes in level, trends, and/or variabilities between phases, demonstrating a functional relation between the paraprofessional-delivered peer training and the focus student's appropriate peer interactions. While Barry had a smaller demonstration of effect due to higher baseline data levels and an upward trend, a functional relation can still be noted.

Other Interaction Variables. Table 2 displays the focus student's independent and prompted initiations and responses with their peer partners and other peers during the baseline and intervention phases. While all focus students increased appropriate interactions with their peer partners, changes were more notable in different measures. For Jacob, he increased his independent initiations and independent responses with his peer partners, from a mean percentage of the interval of 8% to 42% and 11% to 45%, respectively. Jacob was the only focus student with more evident gains in increased communicative turns (from 6% to 31%). With a very small increase in initiations, Max's gain was more evident in his independent and prompted responses toward his peer partners, from 0% to 14% and from 0% to 28%, respectively. Barry's improvement was primarily in his independent responses, which increased from baseline ($M = 39\%$) to intervention ($M = 55\%$). For interactions with other peers, there were very few changes for Jacob and Max between phases. Barry was the only focus student who increased his interactions with other peers, in the areas of independent initiations, independent responses, and appropriate interactions, from an average of 18% to 24%, 19% to 22%, and 18% to 34%, respectively. Across the focus students, there were little to no changes in communicative turns with other peers.

Increased interactions with adults were observed for Jacob and Barry following the intervention. Specifically, Jacob had more independent initiations (19–25%), independent responses (26–31%), and more communicative turns (12–17%). Similarly, Barry also increased his independent initiations (12–30%), independent responses (18–35%), and communicative turns (6%–12%). For adult interactions, there were little to no changes for Max's independent and prompted initiations and responses and for Jacob's and Barry's prompted initiations and responses. For the number of prompts from peer partners, there was a very small increase for Jacob and Max from baseline to intervention (0–1) and no change for Barry (0–0). There was almost little to no difference in the number of different peers interacted between phases, with an average of 3 to 2, 0 to 3, and 5 to 6, for Jacob, Max, and Barry, respectively.

Classroom Variables

Between baseline and intervention phases, the focus student was consistently present during classroom observations with a mean percentage of intervals of 98% to 98%, 95% to 95%, and 93% to 88% for Jacob, Max, and Barry, respectively. In the history class, Jacob received primarily large-group (M = 37% and 46%) and independent work instruction (M = 54% and 31%) during baseline and intervention, respectively. For Max's food science class, the instruction he received during baseline was mostly independent work (56%) and then primarily small-group instruction (62%) during the intervention. For Barry, he received mainly small-group instruction (M = 65% and 72%) followed by no instruction (M = 34% and 27%) across phases.

Across the baseline and intervention phases, both Jacob and Barry were in proximity to at least one of their peer partners for more than 90% of intervals. Max's proximity to his peer partners increased from baseline (M = 31%) to intervention (M = 87%). For proximity to other peers, there was a slight decrease for Jacob (72–67%) and Max (80–74%) and an increase for Barry (78–89%) from baseline to intervention. For proximity to the paraprofessional, there was a slight decrease for Jacob (70–64%), and an increase for Max (79–86%) and Barry (26–39%) from baseline to intervention.

Reliability and Fidelity

Across baseline and intervention observations, we collected IOA for 27%, 23%, and 33% of the sessions for Jacob, Max, and Barry, respectively. Across the students and classes, the average IOA was 99 % (range: 97% to 100%) for all interaction measures (initiations, responses, three or more turns, inappropriate interaction) with peer partners, 100% (range: 98% to 100%) with other peers, and 99% (range: 94% to 100%) with adults. The two observers reached an average IOA of 98% (range: 82% to 100%) for instructional formats, and 99% (range: 97% to 100%) for proximity to peer partners, other peers, and the paraprofessional. The mean total percentage agreement was 83% (range: 0% to 100%) for the number of prompts from a peer partner and 98% (range: 83% to 100%) for the number of different peers interacted.

An observer was present to evaluate the procedural fidelity for the paraprofessional and peer training sessions. All three paraprofessional training sessions had 100% fidelity with all planned items discussed. For Jacob's history class, Bryan was absent so the paraprofessional delivered the training to Molly first with an 89% procedural fidelity. The paraprofessional missed the final reminder of asking the peers to ask the focus student questions and prompting the focus student to talk to other classmates at least three times throughout the class. The observer informed the paraprofessional about the missing step and the paraprofessional redelivered the intervention to both Bryan and Molly with 100% fidelity in the next session. In history class, the intervention observation did not start until both peer partners received the training. For Max and Barry, the paraprofessional delivered the peer training with 100% fidelity.

Social Validity

Among the focus students, Jacob was the one who provided a clear response when asked about his experience with the study. Jacob indicated that he liked to talk to his peer partners in history class. When the researcher asked about their experience with their peer partners, Max did not respond and Barry smiled. Each of the six peer partners completed a brief individual interview, lasting 3 to 4 min. The paraprofessional, three classroom teachers, and two special educators all completed a written questionnaire. All educators affirmed the significance of the study's goal—to evaluate strategies that paraprofessionals and classmates can use to facilitate communication of students in inclusive classrooms. Angela, the paraprofessional, stated, "I feel that Maslow's hierarchy of needs shows that a sense of belonging is imperative before significant learning can occur." Jacob's history teacher, noted, "Communication is critical for student success." One of the special educators shared, "[It is] vital to enable our students to become effective communicators in their school and community environments."

When asked about the procedures of the intervention, the paraprofessional and the peer partners found the training and information shared to be beneficial. Angela indicated, "I did find the prompting and modeling beneficial because I wanted to make sure peers knew what was being requested of them and the focus students to learn appropriate communication skills." Angela also shared that the duration and the information of the initial 1-hr training with the researcher were sufficient but peer training could have been a little

longer. All peer partners noted that the information shared by the paraprofessional was sufficient. Jacob's peer partner, Molly, responded, "Yeah, [the training] made me really think about what I want to [talk about] with [Jacob and another student with disabilities]. When I'm older, I wanted to work at Easter Seals, too. So, it made me happy that I'm glad I was chosen." Barry's peer partner, Rae, also stated, "Yes, it helped me when I did not know how to respond."

Peer partners and educators also noticed positive outcomes as a result of the study. For example, Bryan noticed changes in Jacob's and his own behaviors, "[Jacob is] engaged in more conversations with others. I'm more comfortable to talk about other things." Max's food science teacher, commented on Max taking more initiative to talk to his peer partners. Barry's food science teacher shared,

> [Rae] has always worked very well with [Barry] prior to the project, but I saw an increase in communication and understanding as well as specific task-oriented direction . . . [Austin] and [Barry] had a fun relationship before starting the peer project but it became a more healthy relationship through the course of the project.

Discussion

For students with DD, peer-mediated interventions are embedded with rich benefits, including expanding communication functions, increasing the knowledge and skills of communication partners, heightening diversity awareness, and fostering meaningful friendships (Chung et al., 2012). This study contributed to the evidence base of peer-mediated and paraprofessional intervention research by targeting high school students in general education classrooms and providing low-intensity training. We enhanced the social validity of peer-mediated interventions by equipping a paraprofessional to effectively and efficiently deliver peer training on communication strategies for students with DD in high school general education classrooms. Below we will discuss our key findings based on the two research questions.

Key Findings

Intervention Effects. We are especially encouraged to see increased peer interactions for the three focus students following low-intensity peer training delivered by the paraprofessional. It is noteworthy that the training designed for both the paraprofessional and the peers was relatively brief with simple directions. The paraprofessional training sessions lasted an average of 50 min while the peer training was 3 to 5 min. We considered these training to be low-intensity in comparison with previous studies where the training was more complex and the total training time was more than 3 hr for school personnel (Carter et al., 2016) and 15 to 30 min for peers (e.g., Huber et al., 2018). However, the strength of the intervention seemed to yield different results across students. Specifically, Jacob increased his independent initiations and responses and Max increased his independent and prompted responses almost exclusively with their peer partners as opposed to with other peers or adults in the classroom. While Barry's data were at higher levels from the start, he decreased his inappropriate interactions as his peer partners redirected his loud vocalizations and hugs to more appropriate interactions. Overall, these findings further support peer partners benefiting from the delivery of peer-mediated interventions (Schaefer et al., 2016), although the intervention appeared more effective with students with low to no interactions during baseline due to either lack of opportunities or prompts. Our data showed an increase in responses from all three students, suggesting that these peer partners—who were already in proximity during baseline—*did* increase their initiations toward the focus students following the brief training. However, we recorded limited increases in prompts provided by the peer partners. This result is similar to the findings of Carter and colleagues (2011), where peer partners were less likely to include students with autism in interactions with other classmates in high school settings. It is possible that high school peer partners feel more comfortable serving as communication partners (asking questions and expanding conversations) instead of as facilitators (prompting interactions with other peers). The peer partners may benefit from additional training on strategies to promote students' interactions with others if this is a primary focus. Even with an increase in peer training time to 10 min, the training would still be shorter than in previous studies (e.g., Huber et al., 2018).

There were variable spillover effects with the other peers and adults. Barry increased his independent initiations and responses with other peers and adults and Jacob increased his independent initiations and responses with adults, while Max had little to no changes. Compared with Jacob and Max, Barry seemed to have a more established social network at school and in the classrooms. Thus, other peers may have had a prior relationship with Barry from previous classes and took interest in replicating the observed interactions between the peer partners and Barry. In addition, we noted some potential influences of classroom variables on students' interactions. In Jacob's history class, a substitute paraprofessional was present during two of the three intervention sessions with the highest level of peer interactions. According to our anecdotal notes. Bryan, the peer partner, took on more of a paraprofessional's role in assisting Jacob on these days (e.g., explaining course content and praising Jacob for asking the teacher a question). In contrast, for Max, the lowest level of peer interactions during intervention was a lesson on making cookies—an instructional activity that sparked excitement and high interest among all

students in the class. Perhaps the peer partners were more focused on engaging in the activities and paid less attention to interacting with Max.

Social Validity Results. Results from social validity measures indicated that the peers have improved skill sets in promoting interactions and redirecting students, built their own confidence interacting with diverse students, and expressed their desire to continue or expand interactions beyond the classroom. Not only do peers with these skills help create a more inclusive school, but they can also foster a more supportive community and workplace environment. The paraprofessional shared that her training was a reasonable amount of time, in a feasible delivery model, and overall effective related to the intended purpose of the study. She did state that the peer training could be a little longer although the peers found the training to be sufficient. Together, our findings showed a low-intensity paraprofessional-delivered peer-mediated intervention can effectively increase peer interactions of students with DD in high school general education classes.

Implications for Practice

This study provides a promising model for practical, efficient, and effective paraprofessional and peer training to promote the social interactions of high school students with DD in inclusive classrooms. We conducted one paraprofessional training session for each of the three classrooms; however, it would be possible to conduct one session with brief follow-up sessions. Special educators can develop a brief handout and review the handout with paraprofessionals and peers during short training sessions, as extra planning time is not always available during the school day. Prior to training, educators should consider peer selection based on sharing interests, existing relationships, appropriate social skills, and peer volunteering (Carter et al., 2016). Our anecdotal notes indicated the paraprofessional and the teachers in this study seemed to select peers who had a higher social status. Peers with higher social status are more social and confident and have more friends, which can in turn provide more social opportunities for their partners with disabilities (Chang & Locke, 2016).

It is evident that peers can become impactful natural supports in students' current and future environments. Informed peers can be skilled trainers and/or effective communication partners who model appropriate behaviors and deliver strategies, and ultimately become lifetime advocates for inclusion. The social validity results further confirm the implications of effective peer training for both students with and without disabilities. In particular, general education teachers indicated they welcomed and wanted to foster natural support in their current and future classrooms. The paraprofessional also indicated she felt confident to implement similar strategies with more peers in the future and the peers stated they would implement the strategies in other classes with different students. We hope this study will encourage more educational teams to involve paraprofessionals and general educators in establishing natural support across school environments. With little effort from school personnel already in place, teams can effectively promote interaction between students with and without disabilities.

Limitations

This study has several limitations to consider for interpreting the results and planning future replications. First, the study could have been strengthened with additional peer-fidelity data. While we collected data on peer partners prompting students to interact with others, we did not record if the peer partners prompted the student to respond to their initiations. This information would have allowed for a better understanding of changes in peer partners' behaviors. Second, we did not collect data on the delivery of paraprofessional feedback to peers during the intervention. This coaching time is important and can serve as a reinforcer to peers' behaviors. However, it can be challenging to accurately measure naturalistic and less-intensive interventions. Third, we collected data on all adult interactions together. Separating paraprofessional from general education teacher data could have shown more differentiated changes in students' adult interactions. Fourth, the dependent variable included independent and prompted initiations and responses. It will require additional investigation to determine the relationship between prompts and spontaneous interactions. Fifth, due to the end of the school year, we only had four intervention data points in the last tier and did not collect maintenance and generalization data. Additional data points could have strengthened the study's evidence.

Future Research

While this study demonstrated the effectiveness of low-intensity paraprofessional training and paraprofessional-delivered peer training, this line of research can be extended to add to the literature base of inclusive practices. Future researchers could evaluate additional measures on the implication of strategies by paraprofessionals and peers. For example, the types of conversations (academic or social) or generalization measures in other inclusive settings (e.g., recess and lunch) could be recorded. They could also train paraprofessionals and peer partners on specific strategies to increase students' involvement with other peers during different instructional formats. Finally, future researchers could evaluate the effects of a special educator training general educators on strategies promoting the social communication skills of students with DD related to

class participation with the academic content and with peers. These are future ideas that could further enhance the results of this study illustrating how low-intensity paraprofessional-delivered peer training can increase the appropriate interactions of students with DD in inclusive high school classrooms.

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